



SCIENCE & TECHNOLOGY

Kenya Institute of Curriculum Development





Essence Statement

Science and Technology is a learning area which engages in the human pursuit to understand the relationships between the living and non-living universe. Science is a discipline that deals with explanations and predictions about nature and the universe while Technology is the application of science to create devices that can solve problems and do tasks.

This learning area builds on the competencies introduced at the lower primary under the learning area of Environmental Activities and equips the learner with pre-requisite skills which are required in Integrated Science and Pre-technical and Pre-career studies at the lower secondary level. These enable learners prepare for Science, Technology, Engineering and Mathematics (STEM) in subsequent levels of education cycle.





General Learning Outcomes



By the end of Upper Primary the learner, should be able to:

1. Interact with the environment for learning and sustainable development
2. Apply digital literacy skills appropriately for communication, learning and enjoyment
3. Appreciate the contribution of science and technology in the provision of innovative solutions
4. Use scientific knowledge to observe, explain the natural world
5. Make functional discoveries that impact individuals and the wider society
6. Use innovative approaches as well as critical thinking and problem solving skills to stimulate scientific inquiry, at the local, national and global levels for lifelong learning





Overview of the Design



The design builds up on Environmental Activities in EYE. It has 6 Strands as outlined below:

1. **Living things:** Sub strands are, Plants and Animals
2. **Interrelationship between living and non-living things:** Sub strands are, Air pollution and Water pollution
3. **Digital Technology:** Sub strands are, Digital devices and Coding
4. **Matter:** Sub strands are, Understanding Matter, Properties of Matter,
5. **Force and Energy:** Sub strands are, Force, Energy and Machines
6. **Earth and Space:** Sub strand is Weather and Sky





Science and Technology in Grade 4 forms a strong foundation for subsequent levels in STEM related disciplines by;

- Developing technological skills in the learner through science
- The complementarity between Indigenous technology knowledge systems and Western technology knowledge systems





Unique Features in Science & Tech learning area



Some of the differences between Grade 4 Science & Tech (CBC) and the Standards 4 Science (8-4-4) include;

1. Living Things:

The content on plants has been enriched by variety of suggested learning experiences and content has been added in Characteristics of animals, Vertebrates and Invertebrates and Care of animals.

The content on animals has been enhanced by adding more learning experiences and values. The entire digestive system and its parts are discussed





2. Environment

A new strand called **Environment** is introduced in place of the topic **Water** to address aspects of environmental sustainability.

3. Matter

As a foundation, it is important for the learner to understand matter. However, Pressure, which was included at standard 4 is pushed to the next higher grade level

4. Projects

Unlike science in Grade 4, Science and Tech has added project work, meant to enhance acquisition of core competences and to make the learning hands on.





5. Digital Technology

Digital Technology is a new strand not previously included in the standard 4 syllabus. This strand aims at inculcating innovativeness, creativity, problem solving and analytical thinking skills in the learner.

The content covered at this level include: Digital Devices and Coding
It also inculcates digital literacy which is a 21st Century Skill and it is also one of the 7 Core competencies in CBC.

It develops the ability to use technology as a tool to reinforce, extend and deepen, discover, master and communicate knowledge and information.





Using Problem Solving as a Pedagogy

In Science & Tech., teachers will teach their learners how to solve problems. However, they need to do more than that.

The suggestion is for teachers to prepare learners to be able to use problem solving as a pedagogical tool.

How?

- We should begin teaching with problems, dilemmas and questions
- Focus attention on ideas and sense making rather than on giving directions and rules
- The style of teaching should be the exact opposite of teaching by telling. The teacher should **NOT** do the explaining





- Develop the belief that learners are capable of learning and making sense of their learning.
- Allow students to wonder why things are, to inquire, to search for solutions and to resolve incongruities

PEDAGOGICAL SHIFT

The Three-part Lesson Format is recommended;

1. The “Before” Phase

- Be sure that learners understand the problem
- Get learners mentally prepared to work on the problem
- Clarify expectations prior to learners working on the problem





2. The “During” Phase

- - Let go but listen attentively
- Cautiously provide hints based on learners’ ways of thinking
- Provide profitable activity for learners who finish quickly

3. The “After” Phase

- Engage learners in productive discussion
- Listen attentively without judging
- Summarize main ideas